Embedded Thermostat with Hysteresis Control

RISC-V Assembly Implementation

Embedded Systems | Real-Time Control | Hardware In the Loop

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Preventing Hardware Chatter in Embedded Control Systems

The Problem: Hardware Chatter

Simple Threshold Control

Temperature fluctuations around threshold cause **rapid on/off cycling**

Mechanical Wear

Relay contacts degrade from constant switching

System Instability

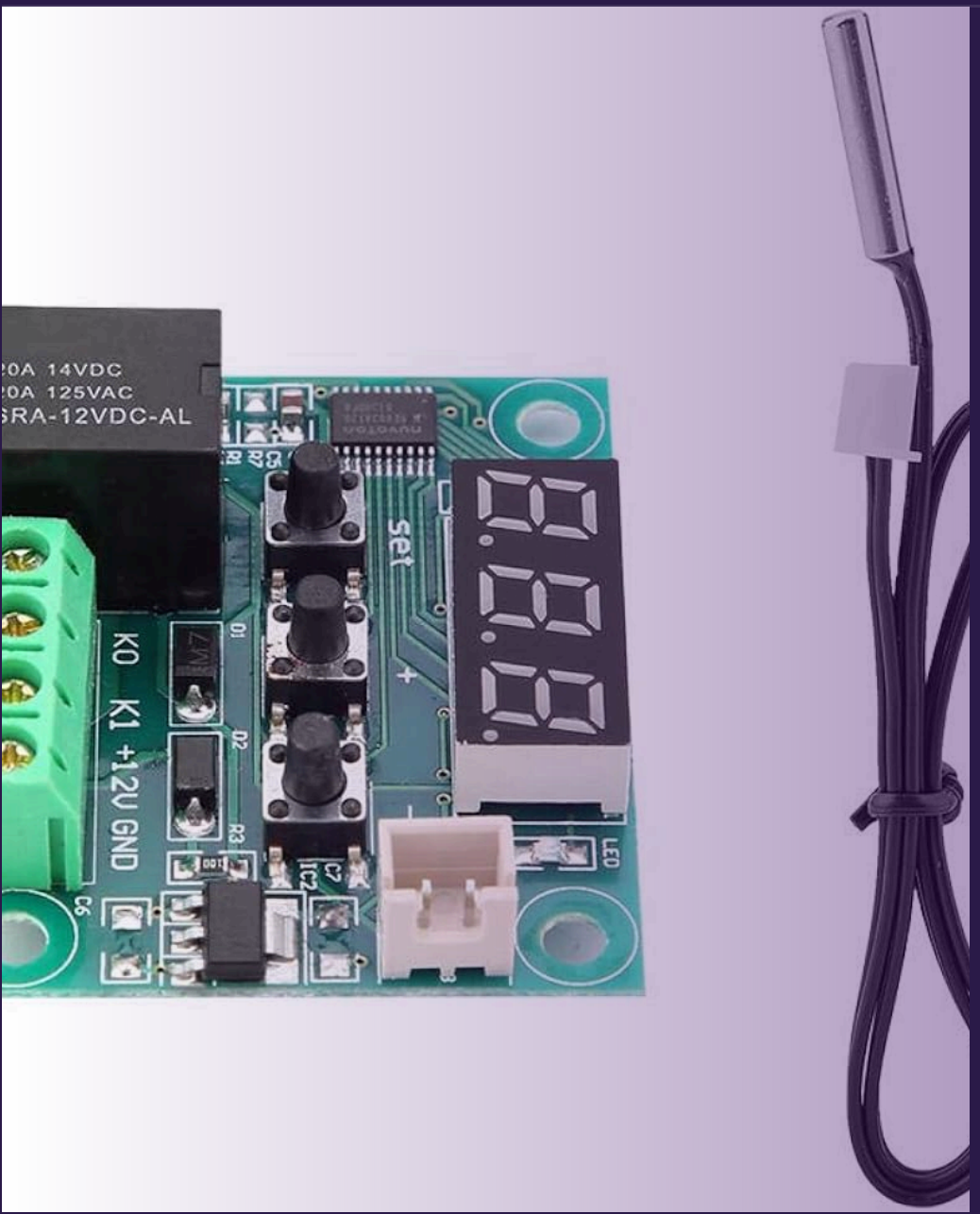
Unpredictable behavior near threshold points

Hardware Damage

Premature failure of electro-mechanical components

Example: Temperature at 20°C → relay switches ON/OFF repeatedly

The Solution: Hysteresis Control



Dead Zone Protection

2°C buffer zone between thresholds prevents oscillation

19°C — Low Threshold

↓ **Turn Heater ON**

Only when temperature drops below

21°C — High Threshold

↑ **Turn Heater OFF**

Only when temperature exceeds

19°C - 21°C: Stable zone - no state changes

System Architecture: Two-State Logic

Finite State Machine

0 **OFF State**

Heater inactive, system monitoring temperature

1 **ON State**

Heater active, warming the environment

Key Insight: State only changes at threshold boundaries

State Transitions

OFF → ON

Temp drops below 19°C

ON → OFF

Temp rises above 21°C

Maintain State

Temp in dead zone 19-21°C

s0 register holds current state

RISC-V Assembly Implementation

Core System Registers

s1 = 19
Low Threshold

s2 = 21
High Threshold

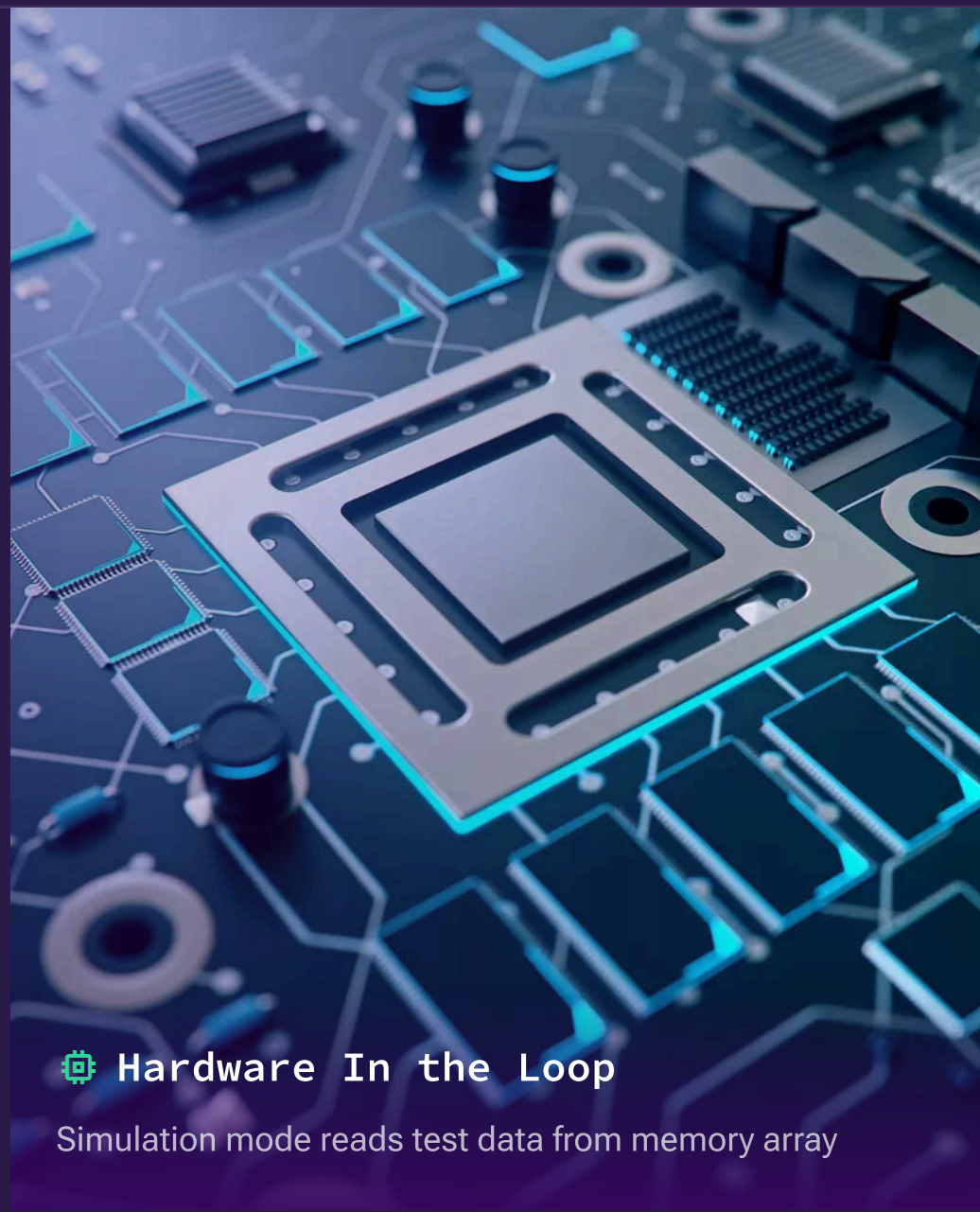
s0 = 0/1
System State

Control Loop Logic

```
lw s3, 0(s4) # Load temperature from memory
```
Sensor data read from simulated stream

```
blt s3, s1, do_turn_on # Temp < 19°C ? Turn ON
```
Check if below low threshold

```
bgt s3, s2, do_turn_off # Temp > 21°C ? Turn OFF
```
Check if above high threshold



Hardware In the Loop

Simulation mode reads test data from memory array

Test Sequence Execution

Temperature Progression

Sensor stream from memory array

22°C → 21°C → 20°C → **19°C** → **18°C** → **20°C** → **22°C**

Heater ON

Temp drops below 19°C

18°C → state = 1

Stable Zone

Dead zone: 19-21°C

No state changes

Heater OFF

Temp exceeds 21°C

22°C → state = 0

Key Result: No rapid switching occurs in dead zone (19-21°C)

Hardware In the Loop

Simulated sensor data validates hysteresis logic

Key Takeaways

Hysteresis Prevents Hardware Damage

Dead zone eliminates rapid switching, protecting relay contacts and extending component lifespan

Real-World Applications

HVAC systems, industrial controllers, automotive thermostats, and embedded heating/cooling control

Hardware In the Loop Testing

Simulated sensor data enables safe validation and debugging before actual hardware deployment

Simple Yet Powerful Implementation

Two-threshold logic provides robust control with minimal code complexity

Essential Pattern: 2°C dead zone between 19°C and 21°C ensures stable operation